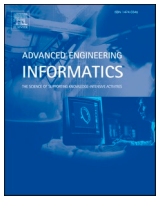
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Full length article



A hybrid CNN-LSTM model for involuntary fall detection using wrist-worn sensors

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ABSTRACT

Falls have become a significant global safety and health concern, which may lead to physical injuries as well as declined mental health, reduced mobility, and deteriorated quality of life. Wearable sensor-based fall detection has emerged as a promising solution for preventing fall-related injuries. However, existing solutions are hindered by user compliance related to sensor placement locations, overall model accuracy, and dependence on simulated voluntary falls. To overcome these limitations, this study aimed to propose a novel involuntary fall detection solution by using wearable sensors and deep learning algorithms. Forty-nine participants were involved in an experimental study, in which activities of daily living and involuntary falls were simulated and kinematic data from these activities were collected using wrist-worn sensors. A novel hybrid model which integrates a convolutional neural network (CNN) and a long short-term memory (LSTM) model was proposed and its performance was compared with the CNN-alone model and LSTM-alone model. The results showed that the proposed hybrid CNN-LSTM model could effectively detect involuntary falls with 96.94% detection accuracy, 98.33% sensitivity, and 96.67% specificity, superior to the CNN-alone model and LSTM-alone model. These results highlight the effectiveness of our proposed approach in significantly improving fall detection accuracy, providing a more reliable and less intrusive solution for preventing fall-related injuries.

1. Introduction

Falls have become a major safety and health concern globally, leading to around 17 million disabilities and 684,000 fatalities annually [1]. Falls not only lead to physical injuries, but also contribute to declined mental health, reduced mobility, and deteriorated quality of life [2]. Fall detection, which can help prompt timely medical attention with little or no human interventions, has been suggested as an effective approach to minimize adverse effects caused by falls and fall-related injuries. Fall detection is considered particularly necessary and important to control the risk of various health complications associated with long lies where fall victims are unable to get up or call for help for an extended period of time [3].

Fall detection technologies can be broadly categorized as context-aware-based or wearable-sensor-based [4]. The context-aware-based fall detection technologies, while useful, are often limited by their confined operational environments, making them suitable primarily for indoor settings. In contrast, wearable-sensor-based fall detection is less restricted by spatial constraints, as it utilizes inertial sensors to detect abrupt body kinematic changes or impacts onto the ground caused by falls, making it portable and versatile. The use of wearable inertial sensors is also more affordable and user-friendly, which can enhance user compliance [5].

Despite the advantages of wearable inertial sensors, their effectiveness has been suggested to be dependent on sensor placement locations [6]. Previous research reported that sensors placed at the lower trunk could lead to desirable detection accuracy [6]. However, this placement can lead to discomfort and even resistance from users, which is a significant barrier to widespread adoption [7,8]. In contrast, researchers have explored the use of wrist-worn sensors, which are less intrusive and more likely to be accepted by users [9,10]. However, the challenge persists because the wrist is often involved in everyday activities presenting similar wrist kinematics with falls (e.g., hand waving), leading to a high incidence of false alarms when detecting falls [11].

To tackle this challenge, there is a growing interest in applying advanced deep learning algorithms in fall detection research. Unlike threshold algorithms and traditional machine learning algorithms that were often seen in wearable-sensor-based fall detection research

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[12,13], deep learning algorithms are good at extracting relevant features reflecting subtle differences between different conditions [14,15]. Therefore, deep learning algorithms have attracted much attention from researchers in recent years. For example, Casilari et al. [16] applied convolutional neural networks (CNNs) for fall detection using multiple publicly available datasets. Similarly, Mauldin et al. [9] proposed a CNN-based fall detection method using smartwatch sensor data. More recently, Alarifi et al. [17] have developed a fall detection framework with the killer heuristic optimized CNN using data from multiple inertial sensors placed on different body parts. Findings from all these studies supported that the CNN-based fall detection model outperformed threshold-based and/or traditional-machine-learning-based models in terms of detection accuracy and models' generalizability. In addition, LSTM models were also of interest because there exist clear temporal patterns in fall dynamics [18], and LSTM is powerful at handling temporal data. Haque et al. [19] conducted experiments comparing three LSTM variants (basic LSTM, CNN-LSTM, and LSTM with self-attention) alongside two Transformer models. Their findings indicated that LSTMs excel in offline fall detection. Al-Qaness et al. [20] presented a deep learning model designed for fall detection that integrates multi-kernel convolution, LSTM, and a self-attention mechanism. By effectively minimizing information loss and improving the differentiation between similar activities, this model achieved high fall detection accuracy rates across several publicly available datasets. However, the current deep learning fall detection models were limited, as they were developed and evaluated using simulated voluntary falls. This raises concerns regarding their applicability to real-world scenarios, where falls are often involuntary and unpredictable.

This study was motivated to overcome the limitations of previous approaches, particularly the high false alarm rate associated with wrist-worn sensors and lack of involuntary fall detection solutions. Specifically, representing a novel contribution to this field, the solution was based on a hybrid model integrating CNN and long short-term memory (LSTM). This hybrid CNN-LSTM model leverages the spatial feature extraction abilities of CNNs alongside the temporal sequence analysis strengths of LSTMs, thereby offering a potential for accurately detecting involuntary falls. At the same time, the proposed hybrid model was trained and assessed using wrist-worn sensor data gathered from involuntary fall scenarios, which could enhance its ecological validity and support its real-life applications.

The contributions and novelties of the present study lie in the following aspects.

- This study developed a novel fall detection solution based on wrist-worn sensors. Wrist-worn sensors are less intrusive and can lead to higher user acceptance, which makes them superior to the most commonly used waist-worn sensors in current fall detection research.
- This study proposed a novel deep learning structure, i.e., CNN-LSTM, to deal with the potential high-false-alarm problem associated with the wrist-worn sensor data. The fall detection model based on this novel structure presented desirable detection performance, suggesting an ideal fall detection solution.
- Involuntary fall data were used in model development and evaluation. The use of involuntary fall data, rather than voluntary fall data as what was done by the existing studies, has taken an important step towards enhancing the ecological validity of the proposed fall detection model and gives us greater confidence that the proposed model is relevant for practical fall detection applications.

2. Methods

Fig. 1 presents an overview of the proposed research. An experiment was conducted to collect data for the development and evaluation of the fall detection model. Forty-nine participants were involved in the experiment (Table 1). They were instructed to perform seven activities of daily living (ADLs), three hand movement activities, and two types of involuntary falls (detailed in Section 2.2), all while wearing a wrist-

Table 1
Participant's demographic information.

	Age (year)			Height (cm)			Weight (kg)		
	Mean	±	std.	Mean	±	std.	Mean	±	std.
Male (N = 28)	24	±	1	175.1	±	5	70.4	±	10.1
Female (N = 21)	23.3	±	1.1	162.1	±	4.6	50.7	±	4.4

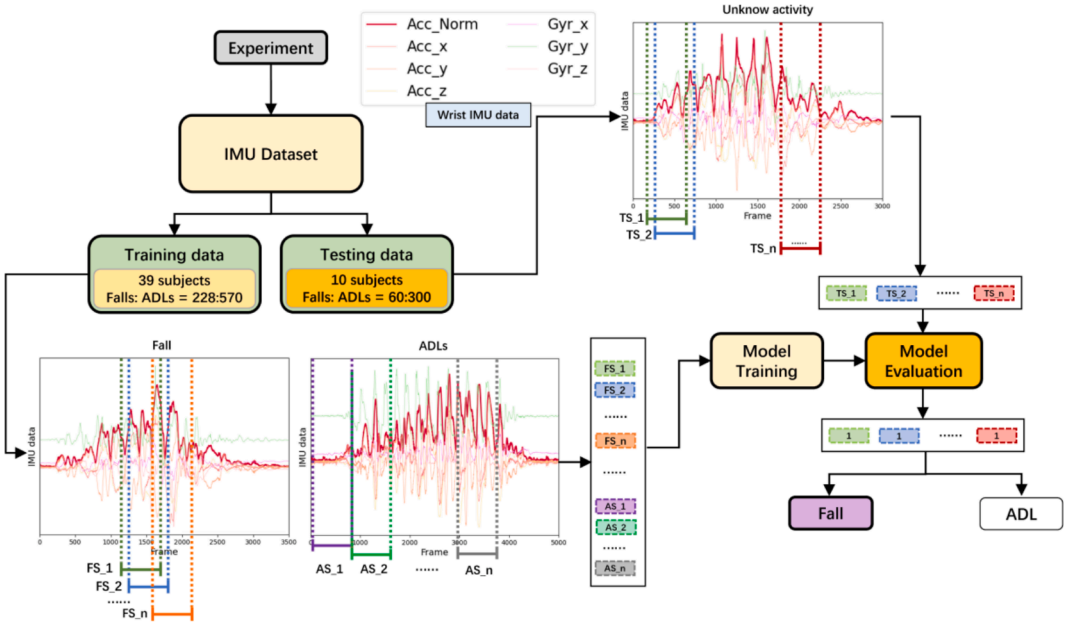


Fig. 1. Overview of the model development and evaluation. FS_1, FS_2,…, and FS_n represent training samples from falling activities; ***AS_1, AS_2,…,*** and ***AS_n*** represent training samples from ADLs. ***TS_1, TS_2,…,*** and ***TS_n*** represent testing samples. t_{Peak} represents the data frame from the peak of the vector sum of the acceleration (*Acc_Norm*).

mounted inertial measurement unit (IMU) (detailed in Section 2.1). The IMU data from 39 participants were used as the training data, and the data from the remaining participants were used for model evaluation. To minimize overfitting effects, a sliding-window data augmentation method was applied to expand the training dataset (Fig. 1). Three different window sizes (0.5 s, 1.0 s, and 1.5 s) were tested (detailed in section 2.3). The model inputs consisted of sequential sliding-window IMU data, while the outputs indicated whether the input was more likely associated with a fall or a non-fall activity. Specifically, the model output "0″ suggested the input originated from an ADL or hand movement activity, whereas "1″ indicated a potential fall. The decision rule for fall detection was based on consecutive outputs: a sequence was classified as a fall if the model produced N consecutive positive values ("1″). Conversely, if the sequence did not meet this criterion, it would be categorized as a non-fall activity. In this study, the performance of the proposed fall detection models was evaluated with different values of N (N = 7, 8, 9, and 10) to identify the optimal threshold for achieving the best fall detection performance.

2.1. Wrist-worn sensor

A customized IMU with the size of 32 mm × 42 mm × 3.5 mm was worn by each participant on the right wrist with its Y-axis aligned with the longitudinal axis of the forearm (Fig. 2). Each IMU contains a 6-channel inertial module (ASM330LXH, STMicroelectronics, Switzerland) that can sense 3D acceleration (±16 g) and 3D angular velocity (±4000°/s). The IMU was equipped with a microcontroller (STM32F103, ST Electronics Ltd, Switzerland) for data logging and storage, and a Bluetooth low energy (BLE) part (Silicon Labs, US) for receiving commands (such as "start the data logging" and "end the data logging") wirelessly from a customized Android App. The IMU was powered by a 3.7 V-230 mAh polymer Lithium-ion battery, and its weight is 8.6 g (with battery). The IMU collected data with a data output rate of 416 Hz.

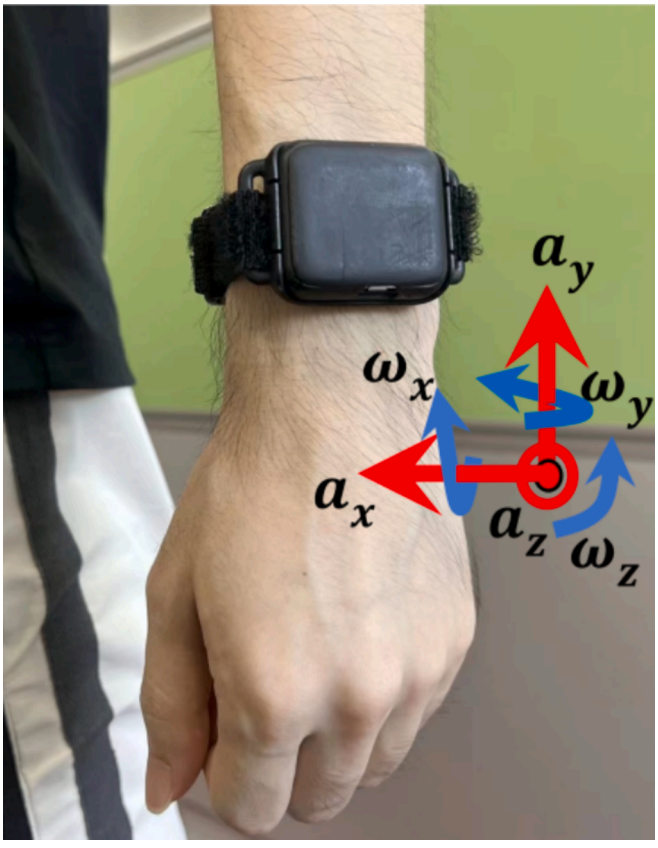


Fig. 2. A customized IMU was worn by each participant with its Y-axis aligned alongside the longitudinal axis of the forearm.

2.2. Experiment

Before the experiment, all the participants signed the informed consent letter. This study aligned with the Declaration of Helsinki and received approval from the University Institutional Review Board.

During the experiment, the participants were instructed to carry out seven ADLs (stair ascent, stair descent, sitting down, standing up, lying down, getting up, and jumping), three hand movement activities (padding, clapping, and waving), and two types of involuntary falls (slip-induced falls and trip-induced falls) during the experiment. The stair ascent and descent tasks were conducted on a 12-step staircase (riser = 18 cm, width = 120 cm, tread = 18 cm). The sit-down/get-up tasks were performed using a wooden chair (height = 49 cm), while the lie-down and get-up tasks took place on a wooden bed with a mattress (length = 200 cm, width = 120 cm, height = 45 cm). For the jumping tasks, the participants were asked to jump vertically as high as possible. In the hand movement tasks, the participants were asked to perform hand waving, clapping, and patting on a desk in a natural manner. Each task was repeated ten times to ensure sufficient data collection. All the ADLs and hand movement activities were performed at participants' self-selected paces.

During the collection of involuntary fall data, participants were instructed to walk on a raised walking platform (12 m × 4 m) along the long-side direction at their self-selected speed, during which slips and trips were randomly introduced. Slips were triggered using a vinyl tile sheet (removable, length = 1.5 m, width = 1.2 m) that was coated with a mixture of 85 % laundry detergent and 15 % water. Trips were triggered by positioning a metal pole 13 cm above the ground. To mitigate the risk of injury from falls, participants were required to wear a full-body safety harness, which was attached to attached to a bearing on a ceiling-mounted track. The harness was adjusted for each participant so that so that their body would be suspended approximately 5 cm above the ground in the event of a complete loss of balance. To avoid anticipation of the falls, the removable vinyl tile was randomly placed at one of three designated locations, approximately 3 to 5 m from the starting point, while the metal pole was randomly placed at one of three potential locations, approximately 6–7 m away from the starting point along the walking path. Additionally, the lighting in the environment was dimmed, and participants were provided with customized sunglasses that obscured the lower part of their visual field when looking straight ahead. This design aimed to take the participants' attention away from the floor surface and make the occurrence of trips and slips unexpected.

A fall was considered to be successfully triggered if the participants' body was held up by the safety harness, and they self-reported their complete loss of balance. Three slip-induced falls and three trip-induced falls were collected from each participant in the experiment. Some experimental trials were excluded from analysis due to sensor noise or data loss. In total, data from 234 involuntary fall trials, 609 ADL trials, and 261 hand movement trials were finally included for model development and evaluation (Table 2).

2.3. Data processing

The raw IMU data from the simulated activities were preprocessed prior to model training. First, the data were filtered using a zero-lag, 2nd-order Butterworth filter with a cutoff frequency of 7 Hz. This helped remove high-frequency noise while preserving the relevant signal characteristics. Next, the filtered data were normalized using the z-score method. This ensured that the data from the accelerometer and gyroscope were on the same scale.

To augment the training data, a sliding-window approach was used. For involuntary falls, the IMU data frame corresponding to the peak acceleration (denoted as t_{Peak}) was first identified, as this frame was assumed to capture the body impact during the fall [21]. The training

Table 2
The number of trials collected for model development and evaluation from each activity.

Type	Activities	Number of trails.
Involuntary Falls	Slip	117
	Trip	117
ADLs	Stair Ascent	87
	Stair Descent	87
	Sit-down	87
	Stand-up	87
	Lie-down	87
	Get-up	87
	Jumping	87
Hand Movements	Padding	87
	Clapping	87
	Waving	87

data were then extracted by applying sliding windows around t_{Peak}, specifically within the range $[t_{Peak}-\frac{1}{2}t_{sliding\ window},\ t_{Peak}+\frac{1}{2}t_{sliding\ window}]$, where $t_{sliding\ window}$ (= 0.5 s, 1.0 s, or 1.5 s) was the size of the sliding window (Fig. 1). The sliding interval was set at 10 % of the window size. For non-fall activities, the training sample data were obtained by sliding windows within the whole period of the simulated activities, maintaining the same window sizes as those used for involuntary falls. To address the significant disparities in the number of trials between involuntary falls and non-fall activities (as shown in Table 2), the sliding interval for non-fall activities was adjusted to be the same as the window size. This adjustment helped mitigate the effects of imbalanced training samples between the two categories.

2.4. Deep learning models

2.4.1. Hybrid CNN-LSTM model

The architecture of the hybrid CNN-LSTM model is depicted in Fig. 3 (a). The CNN module in this model comprises two feature extraction blocks (FEBs). Each FEB consists of a convolutional layer (Conv), a linear rectifier unit layer (ReLU), and a max-pooling layer (MP) (Fig. 3b). The convolutional layer maps the input into features through the convolution operation by the kernels (i.e., a grid of optimizable parameters). The resulting features are activated by the ReLU. The max-pooling layer performs the down-sampling operation which lowers the dimensionality of the resulted features, and subsequently reducing the number of subsequent trainable parameters. Notably, as the CNN module is good at generating fewer complex features than the original input data [22], it is here used to learn and extract features from the input data [23].

For computational efficiency, the LSTM module is designed with a single LSTM layer, followed by a regularization layer (Dropout), two connection regularization Blocks (CRB), a fully connected layer (FC), and a Softmax layer. The cell structure in the LSTM (Fig. 3c) is used to realize the gated mechanism, which is composed of an input gate, a forget gate, and an output gate. With such a gated structure, the previous information can be stored and used to process the current input. Thus, LTSM can handle the vanishing gradient problem by remembering long-term dependencies in the sequential data [24]. Each CRB consists of a fully connected (FC) layer, a linear rectifier unit layer (ReLU), and a

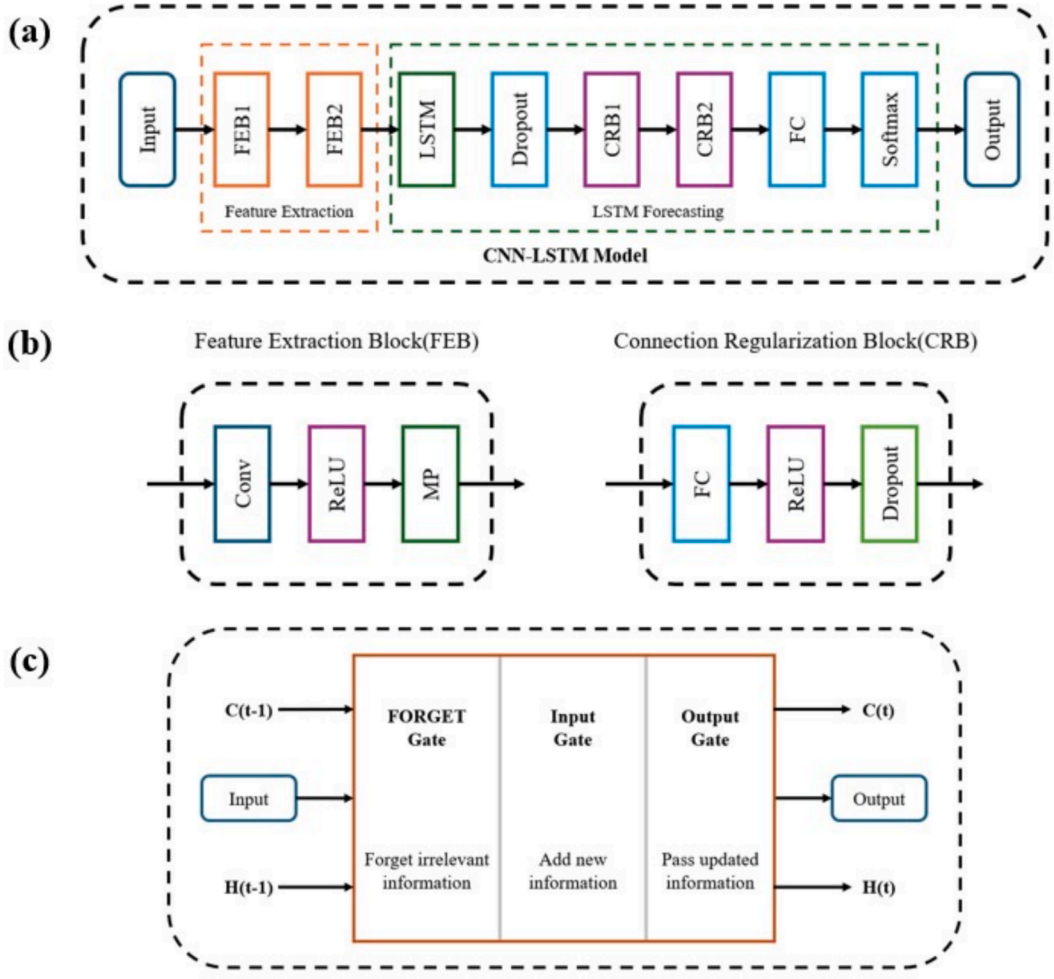


Fig. 3. (a) the CNN-LSTM model; (b) the feature extraction block (FEB) and connect regularization block (CRB); (c) The LSTM cell structure.

regularization layer (Dropout) (Fig. 3b). The key component in the CRB is the Dropout, which is used to create a "thinned" network (i.e., units in the hidden layers are randomly deactivated at various times during training), and minimize overfitting issues with the model. The second CRB is followed by a single fully connected linear layer that maps the extracted features into the final output. Finally, the probabilities of being an ADL or being a fall are calculated by a Softmax layer. In particular, the activity will be concluded as a fall if the probability of being a fall is higher than that of being an ADL. Otherwise, it will be concluded as an ADL.

We also conducted ablation tests to evaluate the contribution of each module to the overall model performance. The tests were conducted in two rounds, with the first round focusing on evaluating the model's performance without the LSTM module, and the second round focusing on evaluating the model's performance without the CNN module.

2.4.2. Comparison models

In addition to the proposed hybrid CNN-LSTM model, two additional models were also used for comparison purpose. These two models were separately based on CNN and LSTM. The CNN-alone model consists of two FEBs, two CRBs, a flattening layer (Flatten), a FC layer, and a Softmax layer (Fig. 4a). The LSTM-alone model consists of two LSTM layers, two Dropouts, three FC layers, two ReLUs, and a Softmax layer (Fig. 4b). Previous studies have demonstrated that two LSTM layers are enough to detect complex features from multi-channel sensor data [25]. The Dropout layer that is added after each LSTM layer can randomly set the outputs of the previous layer to zero and send the scaled outputs to the FC layer selectively activated by the ReLU. Note that the comparative CNN-alone model and LSTM-alone model were respectively different from the ablation models without the LSTM module and without the CNN module, as the CNN– alone and LSTM-alone models were designed to have more complete network architectures.

2.4.3. Model parameters

Tables 3, 4, and 5 summarize the characteristics and hyper-parameters of the utilized models. The model training was carried out with the Adam optimizer [26]. A learning rate decay technique was used during training to avoid overfitting, with the initial learning rate set at 0.001 [27]. The dropout, batch size, and epoch were set at 0.2, 32, and 200, respectively, based on the validation dataset [28]. Cross-entropy was used as the loss function [29]. The model training and evaluation were realized by a customized Python script using Pycharm (version 2020.2). The scripts were running on a computer with a 2.50 GHz Intel i5-7300HQ CPU and 8G RAM. The exemplary data and deployable code are available at https://gitee.com/fang-meng/fall-detection.

Table 3
Characteristics and hyper-parameters of the CNN-LSTM model.

Name	Parameters	Values
Conv1	Filters/Kernel_size/Strides	32/3/1
MP1	Pool_size/Strides	2/2
Conv2	Filters/Kernel_size/Strides	64/3/1
MP2	Pool_size/Strides	2/2
LSTM	Units	64
Dropout	Dropping rate	0.2
FC	Units	2

Table 4
Characteristics and hyper-parameters of the CNN model.

Layers	Corresponding Parameters	Values
FEB1-Conv	Filters/Kernel_size/Strides	32/3/1
FEB1-MP	Pool_size/Strides	2/2
FEB2-Conv	Filters/Kernel_size/Strides	64/3/1
FEB2-MP	Pool_size/Strides	2/2
CRB1-FC	Units	64
CRB1-Dropout	Dropping rate	0.2
CRB2-FC	Units	32
CRB2-Dropout	Dropping rate	0.2
FC	Units	2

Table 5
Characteristics and hyper-parameters of the LSTM model.

Layers	Corresponding Parameters	Values
LSTM1	Units	64
Dropout1	Dropping rate	0.2
LSTM2	Units	64
Dropout2	Dropping rate	0.2
FC1	Units	32
FC2	Units	32
FC3	Units	2

2.5. Model evaluation

The models were evaluated by detection accuracy, sensitivity, and specificity, which were defined as follows.

$$Accuracy = (TP + TN)/(TP + TN + FP + FN) \quad (1)$$

$$Sensitivity = TP/(TP + FN) \quad (2)$$

$$Specificity = TN/(TN + FP) \quad (3)$$

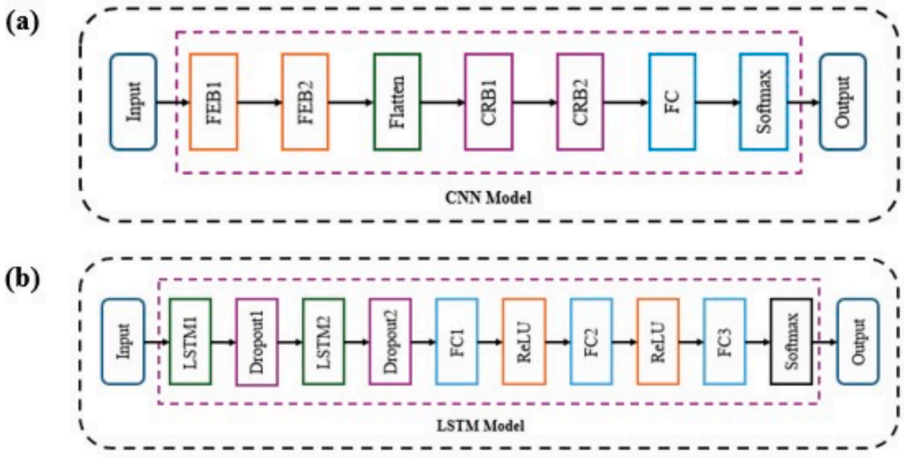


Fig. 4. (a) the CNN model; (b) the LSTM model.

where TP (true positives) represents the number of fall events that were correctly identified by the model, TN (true negatives) represents the number of non-fall activities that were correctly identified by the model, FP (false positives) represents the number of non-fall activities that were mistakenly identified as falls, and FN (false negatives) represents the number of fall events that were mistakenly identified as non-fall activities. Accuracy measures the overall proportion of correct predictions made by models. Sensitivity quantifies models' ability to correctly identify fall events, while specificity measures models' ability to correctly identify non-fall activities. Along with these metrics, a confusion matrix was also presented.

3. Results

The hybrid CNN-LSTM model reached its best detection performance when the sliding window size was set at 1.0 s, and the number of consecutive positive values was 9. Confusion matrix for this optimal model showed that only one slip-induced fall trial was wrongly classified as a non-fall trial, while nine hand-waving and one hand-clapping trials were wrongly classified as falls (Fig. 5). When compared with the CNN-alone model and LSTM-alone model, the CNN-LSTM model achieved the highest detection accuracy (96.94 %) and highest sensitivity (98.33 %) (Table 6). Though the sensitivity of the CNN-alone model was slightly higher (0.33 % higher) than that of the CNN-LSTM model, its sensitivity was 3.33 % lower. The LSTM model appeared to have the worst detection performance, given that it had the lowest values in all three evaluation metrics.

Results of the ablation tests (Table 6) showed that the ablation model without the LSTM module achieved classification accuracy of 96.67 %, matching that of the CNN-alone model, despite minor differences in their layer structures. In contrast, the ablation model without the CNN module attained an accuracy of 94.12 %, which was slightly lower than that of the LSTM-alone model. This is expected, given that the LSTM-alone model incorporated two stacked LSTM cell structures.

4. Discussion

We proposed a novel hybrid CNN-LSTM model for involuntary fall detection in the present study. This model effectively leverages the strengths of both CNN and LSTM architectures, making it capable of extracting distinctive spatial (multi-channel data) and temporal features for falls and non-fall activities. We hypothesized that this hybrid CNN-LSTM model would demonstrate strong performance in fall detection. This hypothesis was supported by our results, with the proposed model achieving a detection accuracy of 96.94 %, sensitivity of 98.33 %, and specificity of 96.67 %. These results also suggested our model to be superior to most of the current wearable-sensor-based deep learning fall detection models. For example, the fall detection accuracies from Hnoohom et al. [30] who proposed a deep residual network with aggregation transformation and from Liu [31] who developed a deep - denoising convolutional autoencoder architecture were 92.52 % and 90.52 %, respectively, which were well below the accuracy obtained by our model.

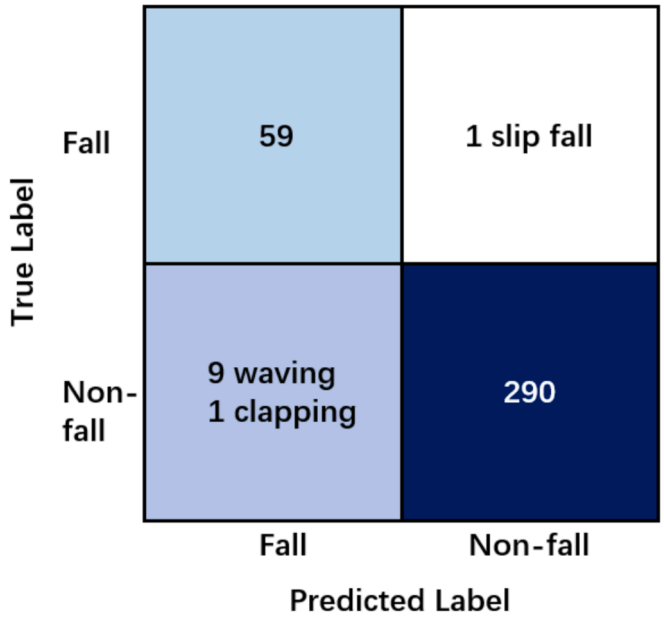


Fig. 5. The confusion matrices of the LSTM model.

Table 6
Model comparison results.

	Accuracy	Sensitivity	Specificity
CNN-alone	96.67 %	95.00 %	97.00 %
LSTM-alone	95.00 %	91.67 %	94.64 %
Ablation without LSTM	96.67 %	95.00 %	97.00 %
Ablation without CNN	94.12 %	93.20 %	94.76 %
CNN-LSTM	**96.94 %**	**98.33 %**	**96.67 %**

In addition, we also tested a CNN-alone model and LSTM-alone model in the present study for comparison purpose. Our findings revealed that the CNN-LSTM model was better than these comparative models. Fall detection in nature is a classification problem. The desirable fall detection performance of the CNN-LSTM model could be attributed to its unique structure, in which the CNN automatically learns and extracts the features characterizing different patterns of the input data, and then the LSTM makes use of the extracted features to accomplish accurate classification [32]. As CNN is capable of processing spatial data while LSTM is powerful for handling temporal data, our finding highlights the importance of both spatial and temporal features in distinguishing falls from non-fall activities. It also implies that an ideal deep learning algorithm for fall detection should be able to recognize both spatial patterns and temporal patterns.

The results from the ablation tests further reinforce the importance of incorporating both CNN and LSTM components in the hybrid model. The ablation model without the LSTM module still achieved a commendable accuracy of 96.67 %, matching that of the CNN-alone model. However, this performance is still poorer than the proposed CNN-LSTM model. The accuracy dropped to 94.12 % when the CNN module was removed, indicating a significant loss in performance. This clearly illustrates the complementary strengths of the CNN and LSTM architectures, where CNN excels in feature extraction from spatial data, and LSTM enhances the model's ability to understand temporal dependencies.

Previous studies also proposed hybrid deep learning frameworks that made use of the advantages of both CNN and LSTM. For instance, Yan et al. [32] presented a CNN-LSTM model for air quality index forecasting, and Ajali-Hernández et al. [33] developed a hybrid deep learning model based on CNN and LSTM for mental task classification. Our proposed CNN-LSTM model is different from these existing CNN-LSTM models given its unique model structure specifically designed for fall detection applications. In particular, our proposed CNN-LSTM model has two feature extraction blocks and two connection regularization blocks in the CNN part. The first feature extraction block can capture low-level, fine-grained features, while the second feature extraction block can learn higher-level, more complex patterns. This hierarchical feature extraction capability enables the model to better represent the intricate spatio-temporal characteristics of the multi-channel IMU data [34]. The inclusion of the two connection regularization blocks helps to regularize the connections between the convolutional layers, promoting robust and generalized feature learning [34]. This design can effectively mitigate overfitting effects, and be beneficial when dealing with limited training data which is always the case for fall

detection applications given the difficulties of collecting fall data.

Another advantage of the proposed model is that it was built on wrist-worn sensors and evaluated by using involuntary fall data. Wrist-worn sensors could lead to high user acceptance, while involuntary falls can better reflect realistic falls. Unlike previous wearable-sensor based fall detection studies that typically placed sensors on the lower trunk and used voluntary fall data for model evaluation, this proposed model is more ecologically valid and more feasible for real life applications. It is also worth noting that once the model has been developed (trained), its deployment does not require significant computational resources. This makes the hybrid model feasible for real-time applications, allowing for efficient integration into practical settings without imposing heavy demands on processing power.

There also exist some fall detection models based on wrist-worn sensors in the current literature. However, these models used either threshold algorithms or traditional machine learning algorithms [6]. The limitation of threshold algorithms is that they cannot account for inter-trial and/or inter-individual differences, which may compromise detection accuracy. At the same time, threshold algorithms are not good at differentiating falls from activities with fall-like movement patterns (e.g., hand clapping) which are often observed from wrist-worn sensor data, thus likely leading to false alarms. Traditional machine learning algorithms need to extract discrete features from wearable sensor time series data. The discrete features may lose important information reflecting subtle differences between falls and non-fall activities and thus compromise detection performance [35]. In contrast to traditional machine learning algorithms, deep learning algorithms can facilitate end-to-end learning to automatically identify subtle features for falls [36]. Many of the fall detection approaches using traditional machine learning reported poor fall detection performance. For example, Mauldin et al.[9] achieved an overall detection accuracy ranging from 65 % to 73 % with Naïve Bayes and support vector machine algorithms, which is far lower than what we reported in the present study.

The simulated activities in the experiment were carefully selected for model development and evaluation. The selected ADLs are about the most common ones in daily life and are associated with a high risk of falls [37]. The selected hand movement activities share similar wrist kinematic patterns with falls. Therefore, they can be used to assess the capability of the proposed models in differentiating falls from fall-like activities. A large proportion of falls are induced by slips and trips [38]. So, we selected slip-induced falls and trip-induced falls for fall detection. It should be noted that the simulated slip-induced falls and trip-induced falls were both involuntary falls. In real life, falls are always involuntary in nature.

This study has some limitations. First, the dataset used for model training and evaluation may not fully account for fall dynamics among the people at high risk of falls, which could limit the model's applicability in fall prevention. To address this, future studies should gather data from high-fall-risk individuals. Second, several factors, such as the height of tripping obstacle and slipping material, were controlled in the experiment, which could limit the models' generalizability. More options for these controlled factors should be examined in future research to improve the model's generalizability. Third, while our study made significant progress by utilizing involuntary fall data, those falls occurred in controlled settings, which may not capture all the complexities of real-world environments where falls typically occur. Future research should prioritize collecting involuntary fall data in natural settings by monitoring fall incidents in everyday situations.

In conclusion, this study has addressed a critical need for effective fall detection solutions by introducing a novel CNN-LSTM model that utilizes wrist-worn sensors. The importance of this work lies in its potential to enhance user acceptance and comfort, as wrist-worn sensors are less intrusive than traditional waist-worn alternatives. By employing involuntary fall data, this research has significantly improved the ecological validity of fall detection models, ensuring them to be more applicable in real-world scenarios. The findings have demonstrated that the proposed model can not only achieve effective fall detection, but also outperform existing deep learning models in this domain. This advancement is crucial for developing reliable fall detection systems, ultimately contributing to improved safety and quality of life for individuals at risk of falls.

CRediT authorship contribution statement

Xinyao Hu: Writing – review & editing, Writing – original draft, Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization. **Shiling Yu:** Visualization, Validation, Software, Formal analysis, Data curation. **Jihan Zheng:** Visualization, Formal analysis, Data curation. **Zhimeng Fang:** Validation, Software, Methodology. **Zhong Zhao:** Writing – review & editing, Validation, Supervision. **Xingda Qu:** Writing – review & editing, Supervision, Resources, Funding acquisition, Conceptualization.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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